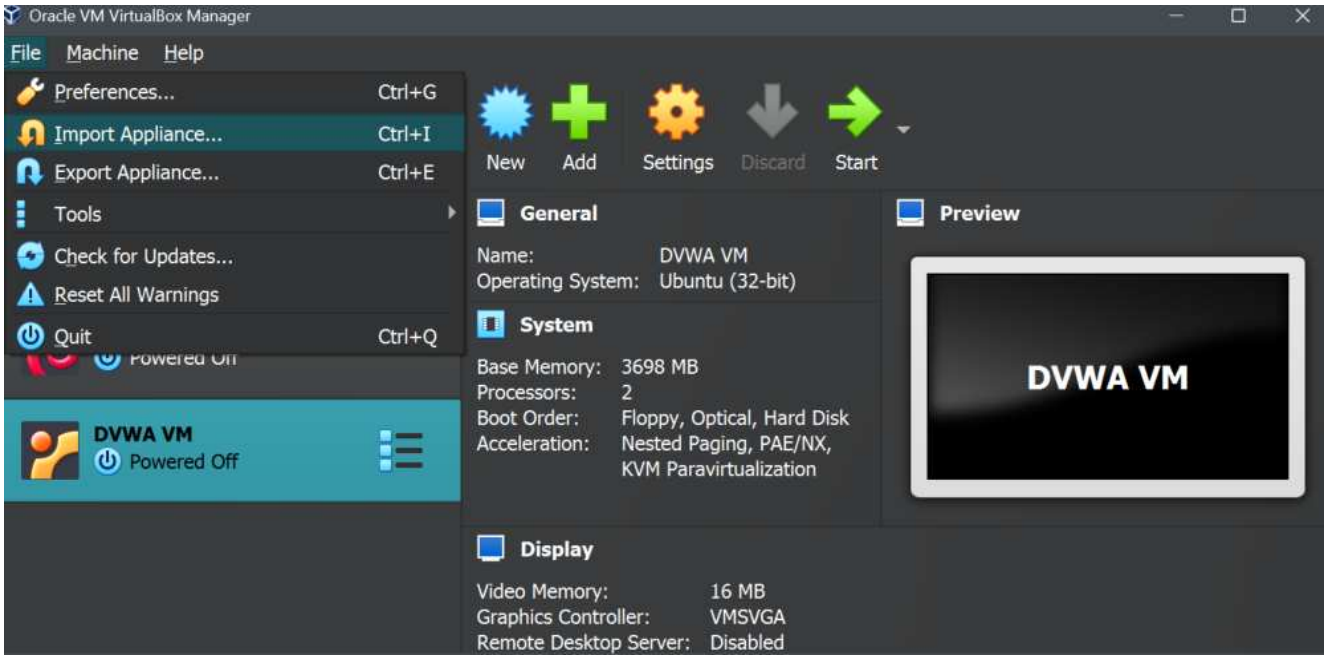
1. INTRODUCTION

- The purpose of this report is to explain, in a simple way, how We carried out forensic investigation on a virtual machine that was provided as a OVA file.
- This training mainly focused on Capturing flags.
- The challenge was to find five hidden “flags,” each placed in different parts of the DF Training server. Each flag required a different type of investigation, such as checking available services, looking through files, decoding hash values, scanning hidden directories, and finally gaining root access to the system.
- The DF training server was first given with no username or password, just an IP address.
- Our final goal was to collect all the flags by working on the IP address using Linux or parrot Virtual Machine and combine those values to get the password for DF server.
- The username was already given and it was also mentioned that the password was a combination of 2 digit code for each flag totalling 8 digits.



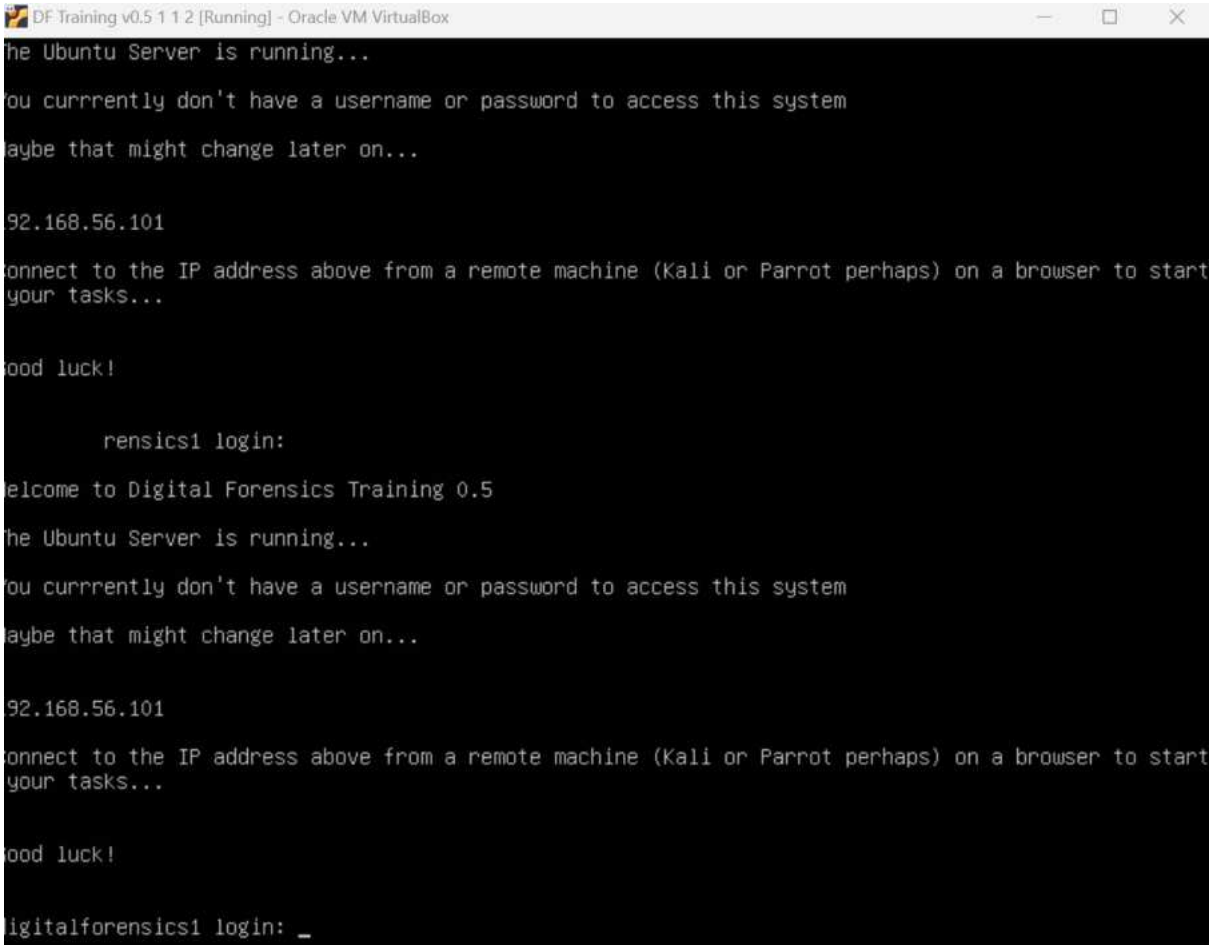
[Image 1 – VirtualBox Import Screen]

[Image 2 – DF Training VM in VirtualBox]

- As displayed in image 1 and 2, We imported the provided OVA file into Virtual Box using Import alliance feature.
- We also ensured that the DF training VM has a Host-only network setting, so that the VM cannot access the internet and internet cannot access the VM.
- This ensures that the investigation is performed in an isolated Environment.

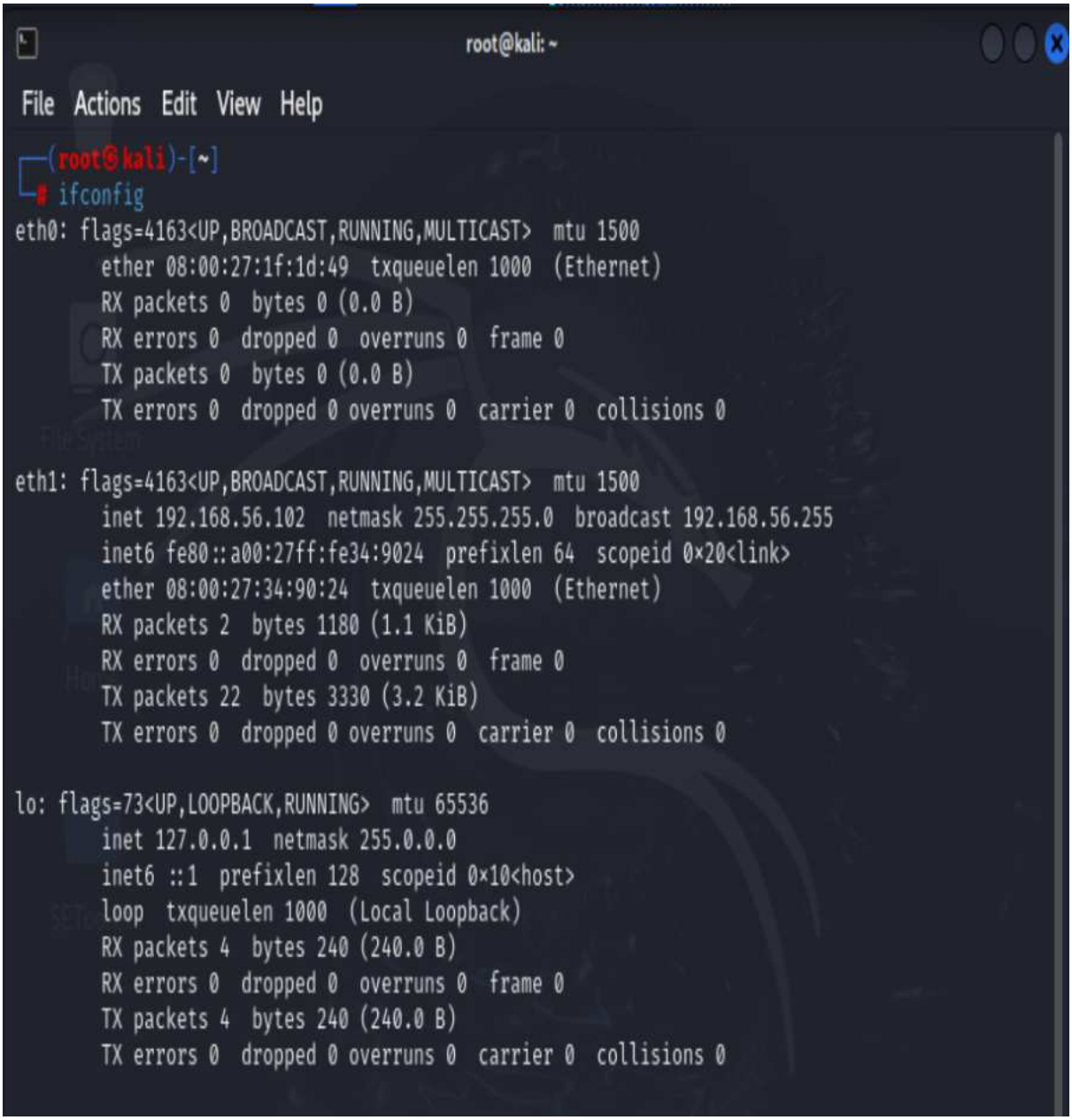
2. ENVIRONMENT SETUP:

- This investigation required two virtual machines in Virtual Box.
    - Kali Linux
    - DF training
- After importing both machines into Virtual Box, we ensured that both machines are set to Host-only network setting, so that the VM cannot access the internet and internet cannot access the VM.
- This ensures that the investigation is performed in an isolated Environment.



[Image 3 – DF Training server page]

- As displayed in Image 3, After booting the DF server it displayed a welcome message and confirmed that ubuntu services are running.
- It also displayed an IP address "**192.168.56.101"** and directed us to connect to the IP address using Linux or Parrot VM to start the tasks.
- The Kali Linux machine was started, and we ensured that both Linux and DF had the same IP address to carry out the tasks efficiently.
- We checked for the Linux IP address using the command "**ifconfig**"



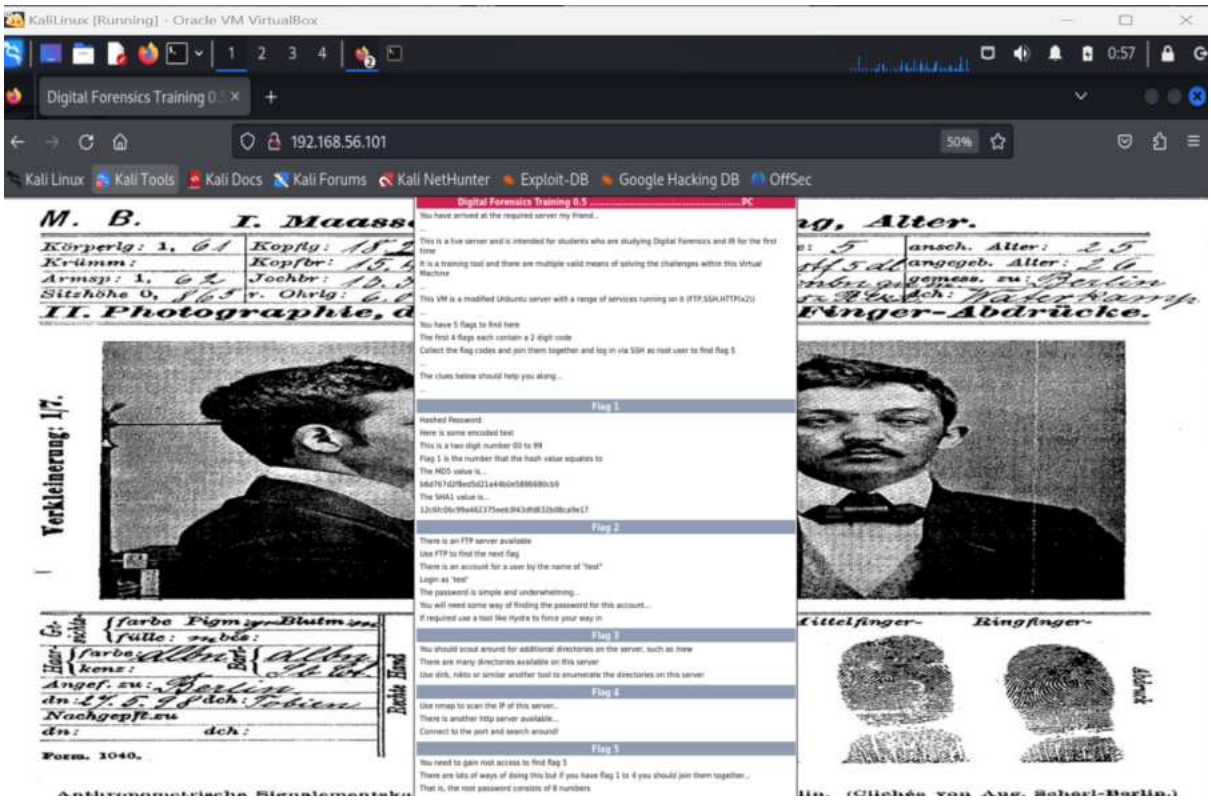
[Image 4 – Kali Linux Network Screen]

- This was done to ensure whether both the machines were operating within same subnet ensuring successful connectivity.
- As shown in image 4, both the machines have same IP ranges.
- After this verification, the IP address was entered into the Linux browser which redirected us to a DF training website.
- This website contained all the tasks that need to be performed to capture all flags.
- It also provided hints on what the flags would be.

- It was mentioned that the first four flags would be two-digit numbers and for accessing the fifth flag we need to combine the first four flags to use it as a password and login to the DF server.

3. CHALLENGE OVERVIEW:

- The challenge introduction page provided the details about the tasks to be completed. The page displayed five flags, each of which would require a different method to discover. The first four were simple two-digit values, while the fifth required logging into the DF Training system using a password which is a combination of the first four flags. The instructions carefully pointed out that the system contained misleading files, suggesting that not every file named as a "flag" should be trusted.
- The challenge mentioned that multiple services would be involved. These included:
    - A web server running on the default port.
    - An FTP server that required weak credentials.
    - Directories that needed brute-forcing because they were not linked anywhere.
    - A second web service running on a non-standard port.



[Image 5 – DF Training Challenge tasks Page]

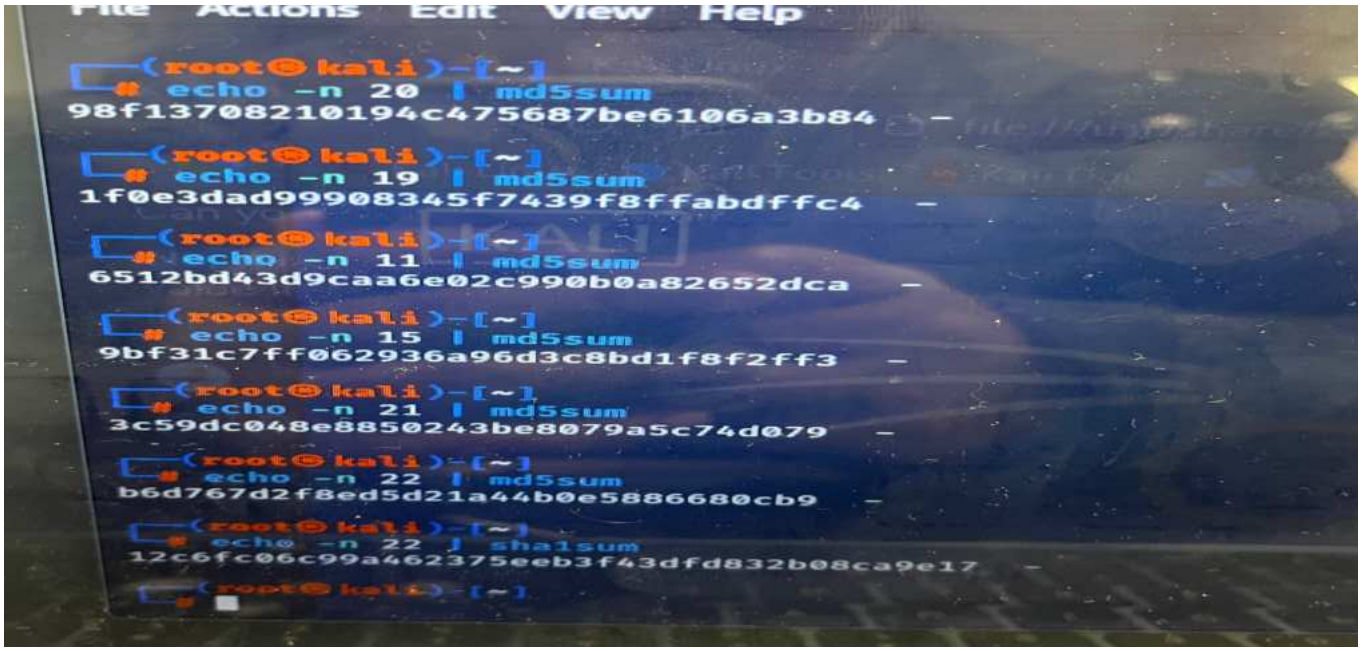
4. Flag 1 – Hashing:

- For the first flag we needed to find the hash value of the give MD5 and SHA1 values.

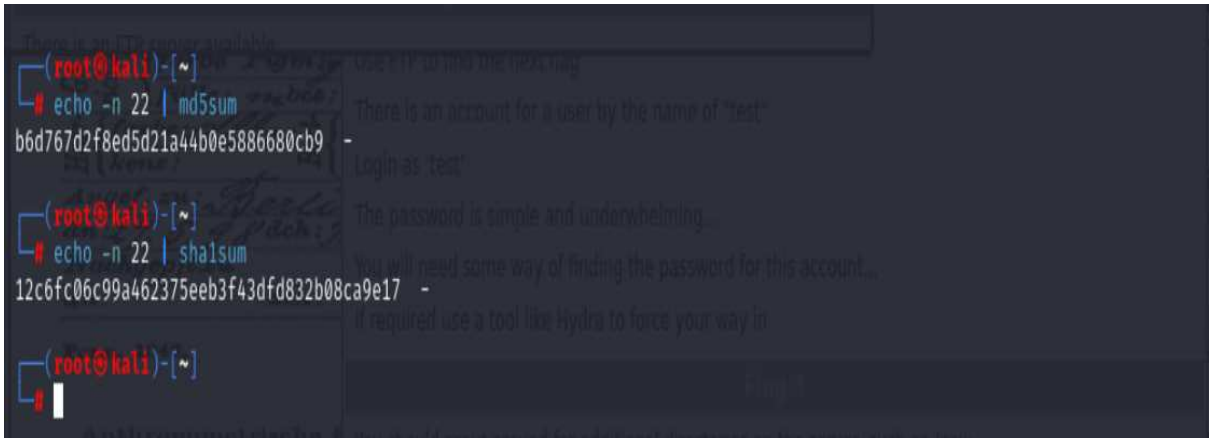
Flag 1
Hashed Password:
Here is some encoded text
This is a two digit number 00 to 99
Flag 1 is the number that the hash value equates to
The MD5 value is...
b6d767d2f8ed5d21a44b0e5886680cb9
The SHA1 value is...
12c6fc06c99a462375eeb3f43dfd832b08ca9e17

[Image 6 – Flag 1]

- The MD5 and SHA1 values given are mentioned in Image 6.
- The approach we used was typing the code
    - **echo -n XX | md5sum**
    - **echo -n XX | sha1sum**



[Image 7 – Flag 1 Brute Forcing]

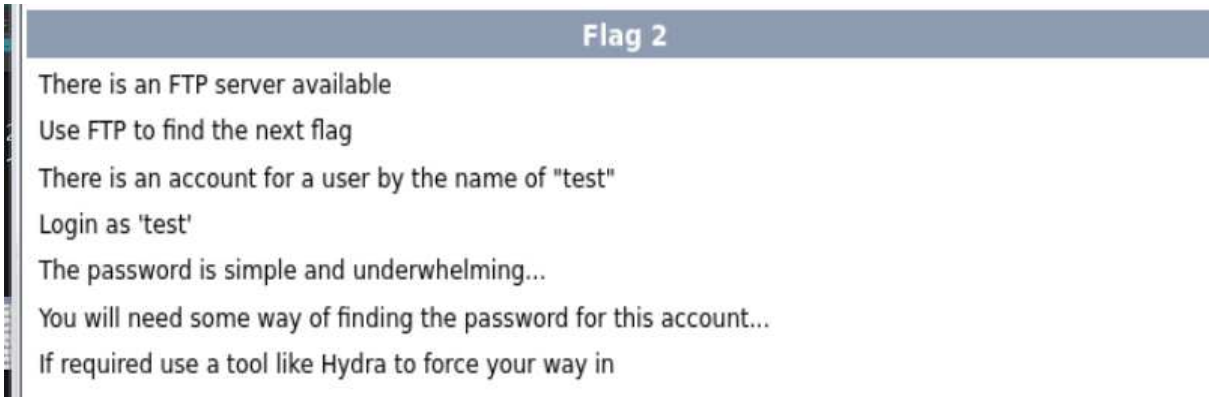


[Image 8 – Flag 1 Result]

- In this code the "**-n**" indicates that there is no newline character in the result, because even a single character can change the result of hash.
- The "**XX**" represents the hash number of the resulting md5 or sha1.
- So we are printing the md5 and sha1 value of the hash number we provide.
- Since it was already mentioned that the value for all four flags are two-digit numbers that is 0-99, we started sequentially entering the numbers starting from **00.**
- After, running the code for several number the hash number "**22**" matched both the md5 and sha1 values.
- With this, the first flag "**22**" was found and documented successfully.

5. Flag 2 – FTP server Investigation:

- For finding the second flag we needed to login to an FTP server using Linux.



[Image 9 – Flag 2]

- It was also clearly given that we needed to used FTP.
- We have been given the username as “test” and password was not given.
- It was mentioned that we need to find password using some way and also use Hydra tool to force our way in.
- So, we moved in to our Linux machine.
- Before using FTP, we need to check whether the TCP port is open In our Linux machine.

```
┌──(root㉿kali)-[~]
└─# nmap ping 192.168.56.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-28 01:21 IST
Failed to resolve "ping".
Nmap scan report for 192.168.56.101
Host is up (0.00037s latency).
Not shown: 997 closed tcp ports (reset)
PORT   STATE SERVICE
21/tcp open  ftp
22/tcp open  ssh
80/tcp open  http
MAC Address: 08:00:27:D7:59:11 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.49 seconds
```

[Image 10 – Nmap scan]

- As shown in image 10, the FTP service is open on port 21, so the port is open and we can successfully log in to FTP server.

```
┌──(root㉿kali)-[~]
└─# ftp 192.168.56.101
Connected to 192.168.56.101.
220 ProFTPD 1.3.5e Server (Debian) [::ffff:192.168.56.101]
Name (192.168.56.101:root): test
331 Password required for test
Password:
230 User test logged in
Remote system type is UNIX.
Using binary mode to transfer files.
ftp>
```

[image 11 – FTP server login]

- Image 11 shows the login of FTP server.
- We entered the ftp command with the IP address of the Machine.
- The username is "**test**" as already given.
- Most of the Servers default password would be either empty or "12345" or the username itself.
- After trying the combinations we assumed, the password was same as the username as expected.
    - **FTP server: Login = test**
    - **Password = test**
- The type of remote system is UNIX
- After successful login into the ftp server, we began the investigation.
- At first we tried to check if there is any files or folders present in the ftp server using the "**ls**" command.

```
ftp> ls
229 Entering Extended Passive Mode (|||49370|)
150 Opening ASCII mode data connection for file list
drwxr-xr-x   2 root     root        4096 Oct 12  2023 directory
drwxr-xr-x   3 root     root        4096 Oct 15  2023 files
-rw-r--r--   1 root     root         268 Oct 15  2023 hint.txt
226 Transfer complete
```

[image 12 – Ftp server Investigation**]**

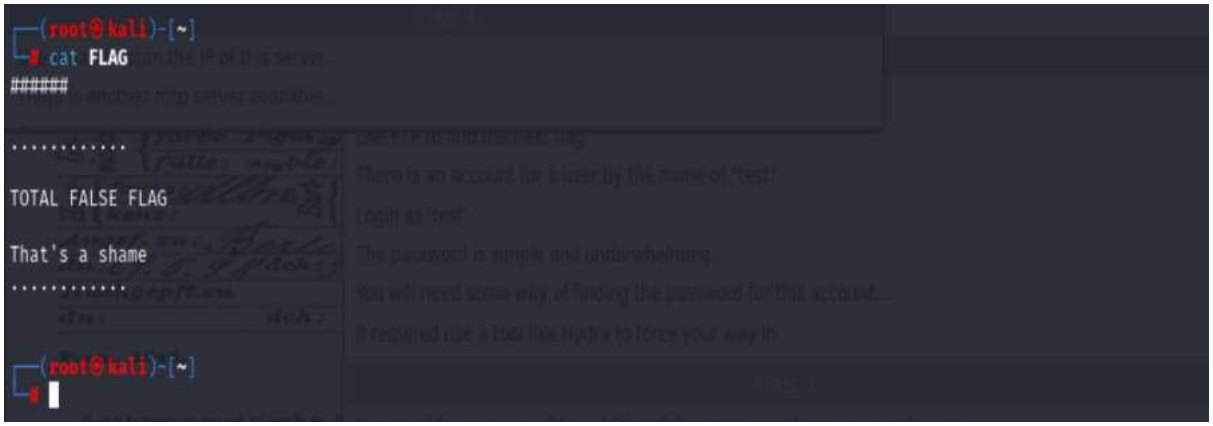
- As shown in image 12 there were 3 items in the server.

- Two were folders and one was a file.
- The first folder was named as **directory** and for it the **"User had read, write and execute permissions", "Groups had read and execute permissions", "Others had only execute permissions".**
- The second folder was named as **files** and for it **"User had read, write and execute permissions", "Groups had read and execute permissions", "Others had only execute permissions".**
- The third file was name as **hint.txt** and for it **"User had read and write permissions", "Groups had only read permissions", "Others had only read permissions".**
- At first we wanted to check for the **directory** folder for any flags.
- The **directory** folder was created on Oct 12 2023.
- We used "**cd**" to go into the directory folder.

```
ftp> cd directory
250 CWD command successful
ftp> ls
229 Entering Extended Passive Mode (|||45846|)
150 Opening ASCII mode data connection for file list
-rw-r--r--   1 root     root           70 Oct 12  2023 FLAG
226 Transfer complete
ftp> get FLAG.txt
local: FLAG.txt remote: FLAG.txt
229 Entering Extended Passive Mode (|||25160|)
550 FLAG.txt: No such file or directory
ftp> get FLA
local: FLA remote: FLA
229 Entering Extended Passive Mode (|||25204|)
550 FLA: No such file or directory
ftp>
ftp>
ftp> get FLAG
local: FLAG remote: FLAG
229 Entering Extended Passive Mode (|||49851|)
150 Opening BINARY mode data connection for FLAG (70 bytes)
100% |***********************************************************************************************|    70       33.80 KiB/s    00:00 ETA
226 Transfer complete
70 bytes received in 00:00 (12.36 KiB/s)
ftp> bye
221 Goodbye.
```

[image 13 – Investigating Folders in FTP]

- The **directory** folder had a file called **"FLAG"**
- We expected this to be the second flag.
- We cannot view the contents of file directly in FTP server so, we used the command **"get FLAG",** like shown in image 13.
- This command stores the **FLAG** file to the root directory of our Linux.
- Then we exit the FTP server using the command **"bye".**
- After ensuring we were in the root folder of Linux we used **"cat FLAG"** to view the contents in the file.



[image 14 – FALSE FLAG]

- But the result was not according to our expectations.
- The file only contained a **"FALSE FLAG",** maybe used as an distraction.
- So, we logged in back to the FTP server with the same command **"ftp 192.168.56.101"** and the same login credentials as used before and continued to the next folder.
- The next folder is a folder named **"files".**
- We continued into "**files"** using "**cd"** command.
- The **files** folder was created on oct 15 2023.
- This folder contained many files name
    - a – z
    - 000 – 100
    - README.txt
    - Info.info
    - FLAG2.txt

```
-rw-r--r--    1 root     root            6 Oct 12  2023 File020.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File021.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File022.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File023.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File024.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File025.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File026.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File027.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File028.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File029.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File030.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File031.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File032.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File033.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File034.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File035.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File036.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File037.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File038.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File039.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File040.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File041.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File042.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File043.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File044.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File045.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File046.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File047.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File048.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File049.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File050.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File051.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File052.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File053.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File054.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File055.txt
```

[image 15 – files directory]

```
drwxr-xr-x    2 root     root         4096 Oct 12  2023 directory
drwxr-xr-x    3 root     root         4096 Oct 15  2023 files
-rw-r--r--    1 root     root          268 Oct 15  2023 hint.txt
226 Transfer complete
ftp> cd files
250 CWD command successful
ftp> ls
229 Entering Extended Passive Mode (|||52013|)
150 Opening ASCII mode data connection for file list
drwxr-xr-x    6 root     root         4096 Oct 12  2023 assets
-rw-r--r--    1 root     root            0 Oct 12  2023 a.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 b.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 c.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 d.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 e.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File000.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File001.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File002.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File003.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File004.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File005.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File006.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File007.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File008.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File009.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File010.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File011.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File012.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File013.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File014.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File015.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File016.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File017.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File018.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File019.txt
```

[image 16 – files directory]

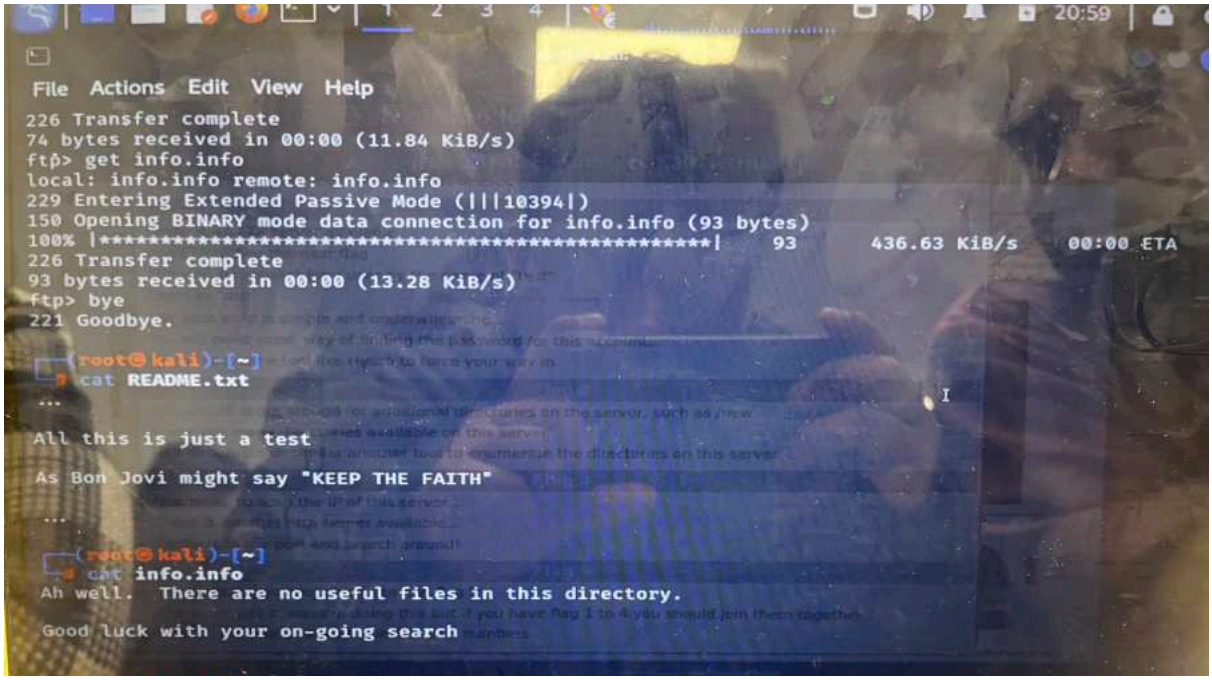
```
-rw-r--r--    1 root     root            6 Oct 12  2023 File054.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File055.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File056.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File057.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File058.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File059.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File060.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File061.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File062.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File063.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File064.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File065.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File066.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File067.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File068.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File069.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File070.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File071.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File072.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File073.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File074.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File075.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File076.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File077.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File078.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File079.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File080.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File081.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File082.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File083.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File084.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File085.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File086.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File087.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File088.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File089.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File090.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File091.txt
```

[image 17 – files directory]

```
-rw-r--r--    1 root     root            6 Oct 12  2023 File090.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File091.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File092.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File093.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File094.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File095.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File096.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File097.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File098.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File099.txt
-rw-r--r--    1 root     root            6 Oct 12  2023 File100.txt
-rw-r--r--    1 root     root           34 Oct 15  2023 FLAG2.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 f.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 g.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 h.txt
-rw-r--r--    1 root     root           93 Oct 12  2023 info.info
-rw-r--r--    1 root     root            0 Oct 12  2023 i.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 j.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 k.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 l.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 m.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 n.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 o.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 p.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 q.txt
-rw-r--r--    1 root     root           74 Oct 12  2023 README.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 r.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 s.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 t.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 u.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 v.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 w.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 x.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 y.txt
-rw-r--r--    1 root     root            0 Oct 12  2023 z.txt
226 Transfer complete
ftp> get FLAG2.txt
local: FLAG2.txt remote: FLAG2.txt
```

[image 18 – files directory]

- The files with names a to z were empty.
- The files named 000 to 100 contained only **“hello”** as a content inside it.
- Each file were checked and everything had the same result.
- The **“README.txt”** file contained , **“All this is just a test, As Bon Jovi might say "KEEP THE FAITH" “.**
- The “**info.info**” file contained, **“Ah well. There are no useful files in this directory. Good luck with your on-going search”.**

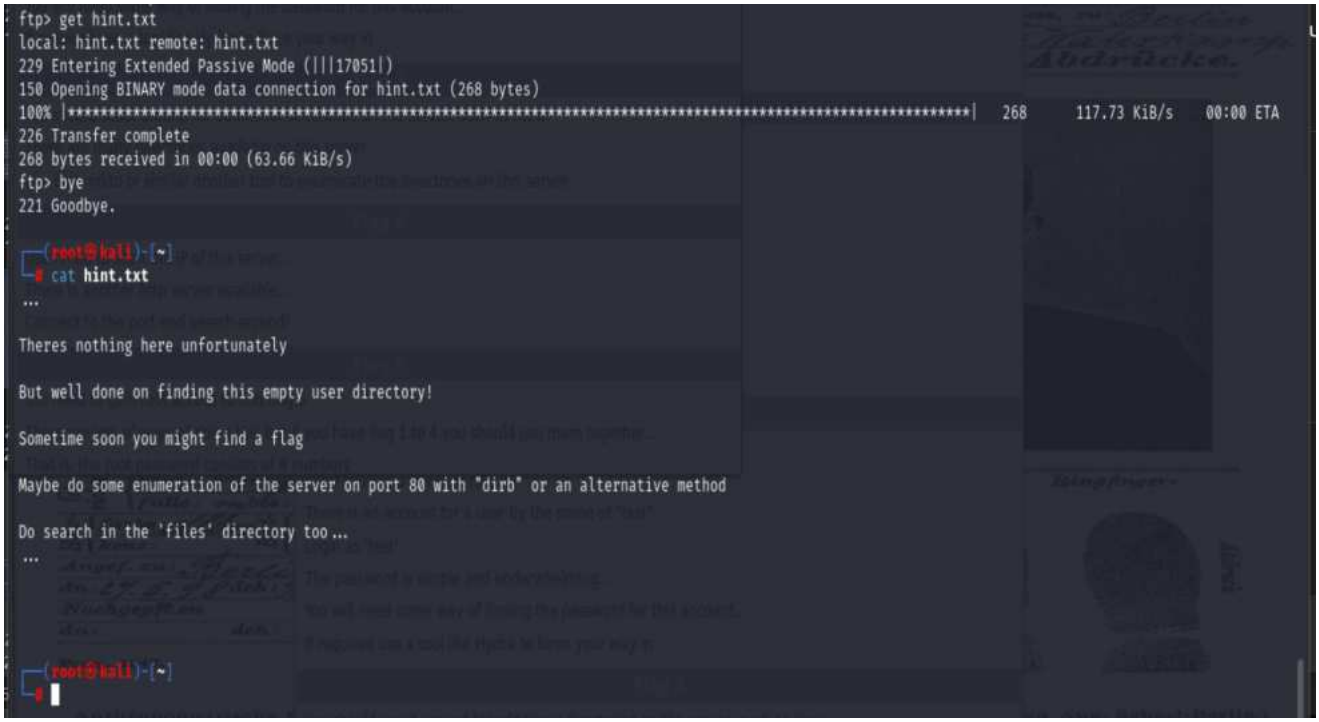


[image 19 – Exploring other files]

- All the files mentioned in image 15 – 20 were imported to Linux machine using **“get”** command, the FTP server was exited using **“bye”** command and the contents were read from the Linux root using **“cat”** command.
- Then we come to the last file in **“files”** directory.
- The file is named as **“FLAG2.txt”**
- Like mentioned before, The file was extracted to root folder, We exited the server and the contents were read.

[image 20 – Flag 2 Result]

- As seen in image 20 **Flag 2** was successfully identified from the **"FLAG2.txt"** folder.
- With this, the second flag "**44**" was found and documented successfully.
- Eventhough we have found the result there was another file in the server named **"hint.txt" created on oct 15 2023.**
- We assumed that this file might have hint for Flag3 so, as we did before the file was extracted and the contents were read.



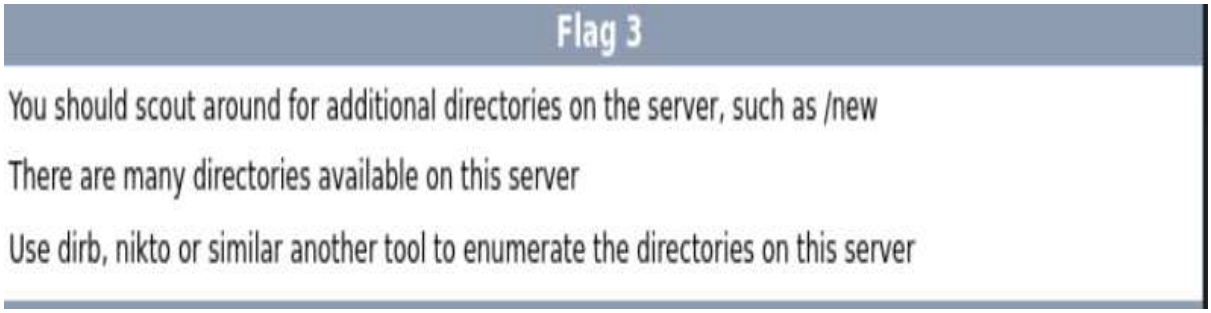
[image 21 - hint]

- As assumed the text file contained an hint for flag 3 (image 21).

- Eventhough it was an empty user directory, the text inside the file suggested us to use **"dirb"** or any other alternate method in **port 80.**
- It also suggested us to search **files** directory which could have been a Flag 2.

6. Flag 3 – Navigating hidden directories in FTP Server:

- We again went to the DF training website to get hints on Flag 3.
- For Flag 3 it was mentioned that further investigation on the FTP server will lead us to the result.
- Hidden directories like /new would be found.
- It is also mentioned that we can use dirb, nikto or any other similar tools to find the hidden directories in the server.



[image 22 – Flag 3]

- As the task suggested we used the dirb command.
- Dirb is best used for scanning hidden folders, whereas nikto is for server vulnerabilities.
- Since, our task is to extract hidden folders we are using dirb.

```
┌──(root㉿kali)-[~]
└─# dirb http://192.168.56.101

-----------------
DIRB v2.22
By The Dark Raver
-----------------

START_TIME: Fri Nov 28 01:50:54 2025
URL_BASE: http://192.168.56.101/
WORDLIST_FILES: /usr/share/dirb/wordlists/common.txt

-----------------

GENERATED WORDS: 4612
---- Scanning URL: http://192.168.56.101/ ----
==> DIRECTORY: http://192.168.56.101/admin/
==> DIRECTORY: http://192.168.56.101/backdoor/
==> DIRECTORY: http://192.168.56.101/data/
==> DIRECTORY: http://192.168.56.101/database/
==> DIRECTORY: http://192.168.56.101/dev/
==> DIRECTORY: http://192.168.56.101/hide/
+ http://192.168.56.101/index.html (CODE:200|SIZE:7601)
==> DIRECTORY: http://192.168.56.101/js/
==> DIRECTORY: http://192.168.56.101/new/
==> DIRECTORY: http://192.168.56.101/php/
==> DIRECTORY: http://192.168.56.101/python/
==> DIRECTORY: http://192.168.56.101/secret/
+ http://192.168.56.101/server-status (CODE:403|SIZE:279)
==> DIRECTORY: http://192.168.56.101/upload/

---- Entering directory: http://192.168.56.101/admin/ ----
(!) WARNING: Directory IS LISTABLE. No need to scan it.
    (Use mode '-w' if you want to scan it anyway)
```

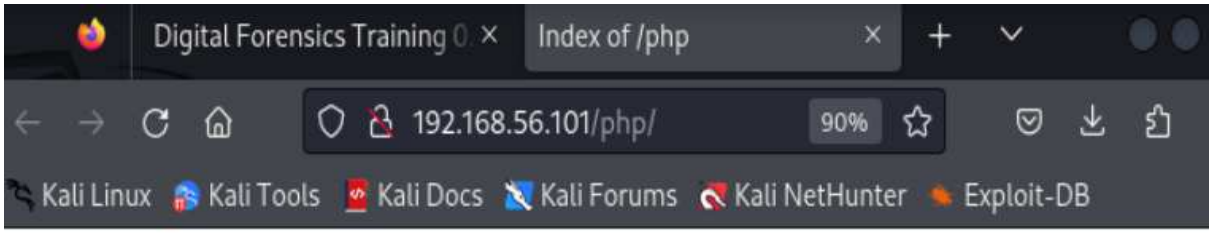
[image 23 – Scanning hidden folders using dirb]

- We used **DIRB with the HTTP protocol and the IP address** because DIRB needs a **web target** to scan. DIRB works by sending requests to a web server and checking which directories or files exist.
- Since our DF training runs only on IP and not a domain we are using the IP with HTTP.
- **DIRB** needs a proper web URL to scan for hidden directories.
- DIRB brute-forced the web server by trying many common folder names and checking which ones returned a valid response.
- After using the Dirb command with IP address, Dirb returned **"12 hidden directories"(**image 23**)**
- We went through each and every URL by pasting it in browser.

```
=> DIRECTORY: http://192.168.56.101/new/
=> DIRECTORY: http://192.168.56.101/php/
```

[image 24 – php folder]

- After trying each and every website in a sequential order, we accessed the folder **"/php/".**
- This folder contained many files but those were empty



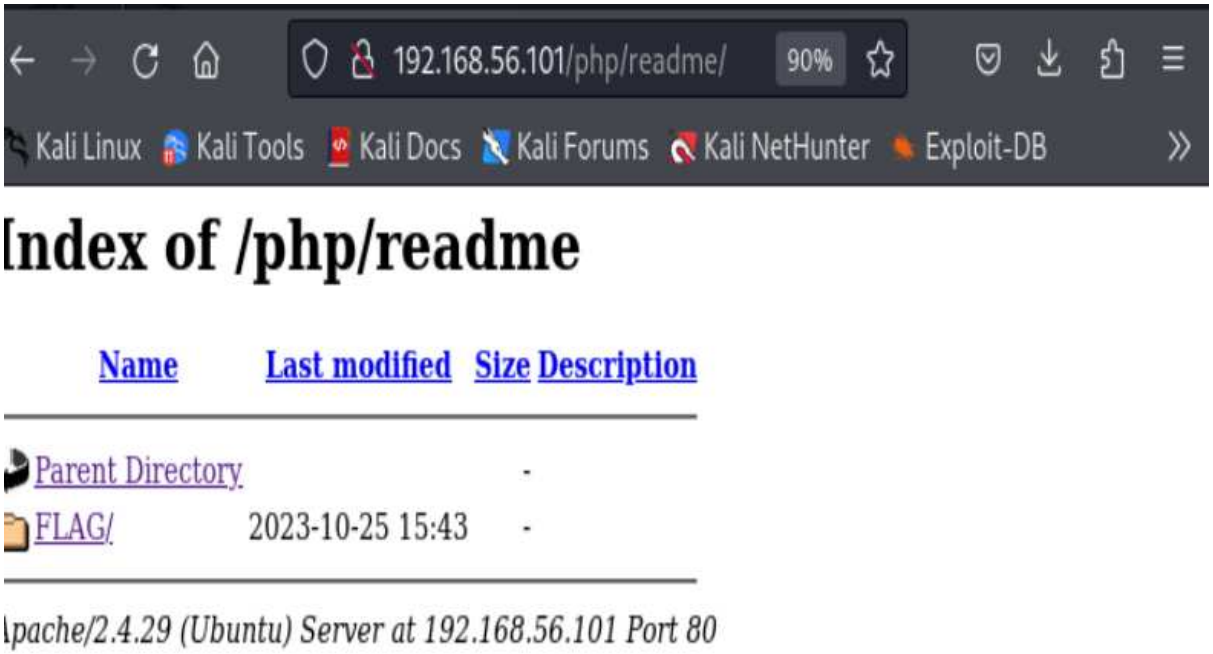
Index of /php

Name	Last modified	Size	Description
Parent Directory		-	
dvwaExecLow.php	2023-09-27 23:54	411	
dvwaExecLow2.php	2023-09-28 00:00	0	
dvwaExecLow3.php	2023-10-16 13:24	581	
nothing.txt	2023-09-27 23:48	36	
pingToolWithCommandInjection.php	2023-09-28 00:16	958	
readme/	2023-10-25 15:43	-	
reverseShell.php	2023-09-29 19:44	5.4K	
test.php	2023-09-27 23:46	20	

Apache/2.4.29 (Ubuntu) Server at 192.168.56.101 Port 80

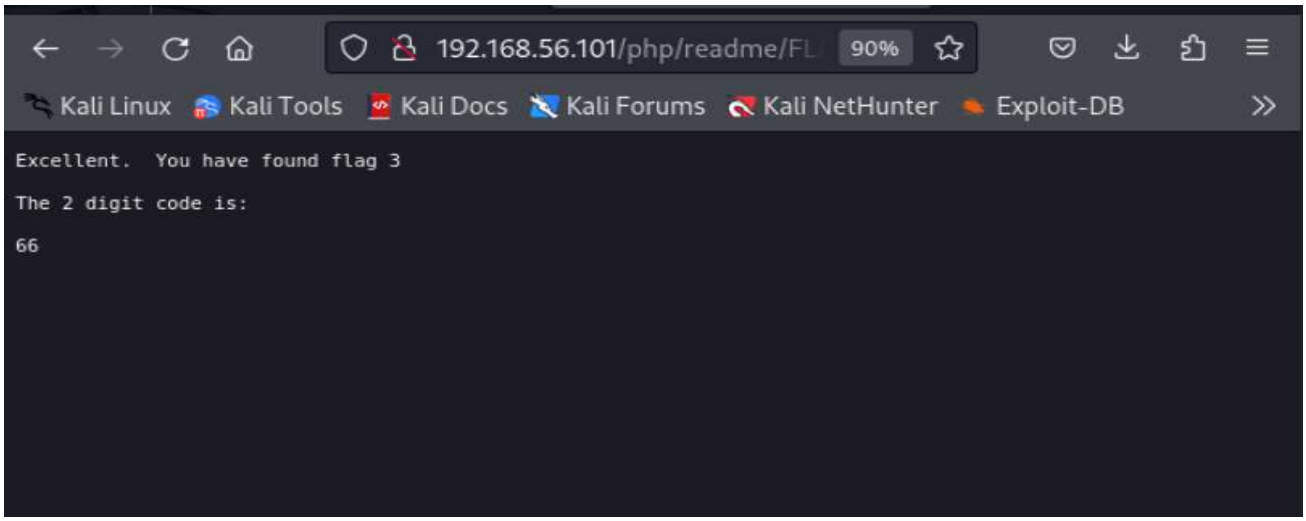
[image 25 – Investigating php folder]

- Another subfolder named **"/php/readme" was present inside /php/(image 25).**



[image 26 – Readme folder containing another folder named flag]

- The readme folder contained another subfolder named **FLAG/(image 26).**
- Finally inside the FLAG/ folder there was file named **"flag3.txt"**
- The final path was **/php/readme/FLAG/**



[image 27 – Flag 3 Result]

- With this, the third flag "**66**" was found and documented successfully(image 27).

- **The flags that have been found till now is 22,44,66.**

7. Flag 4 – Finding and Investigating Http server:

- Flag 4 mentions that we should scan the IP address of the server using **"nmap".**
- It is also mentioned that another http server is running in a different port.
- Our task is to find the port and search around it for fourth flag.

Flag 4

Use nmap to scan the IP of this server...

There is another http server available...

Connect to the port and search around!

[image 28 – Flag 4]

- To find the hidden port we should do a full port scan using nmap.

```
┌──(root㉿kali)-[~]
└─# nmap -p- 192.168.56.101
Starting Nmap 7.94SVN ( https://nmap.org ) at 2025-11-28 02:05 IST
Nmap scan report for 192.168.56.101
Host is up (0.00039s latency).
Not shown: 65531 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
22/tcp    open  ssh
80/tcp    open  http
14545/tcp open  unknown
MAC Address: 08:00:27:D7:59:11 (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 38.85 seconds
```

[image 29 – Full port scan]

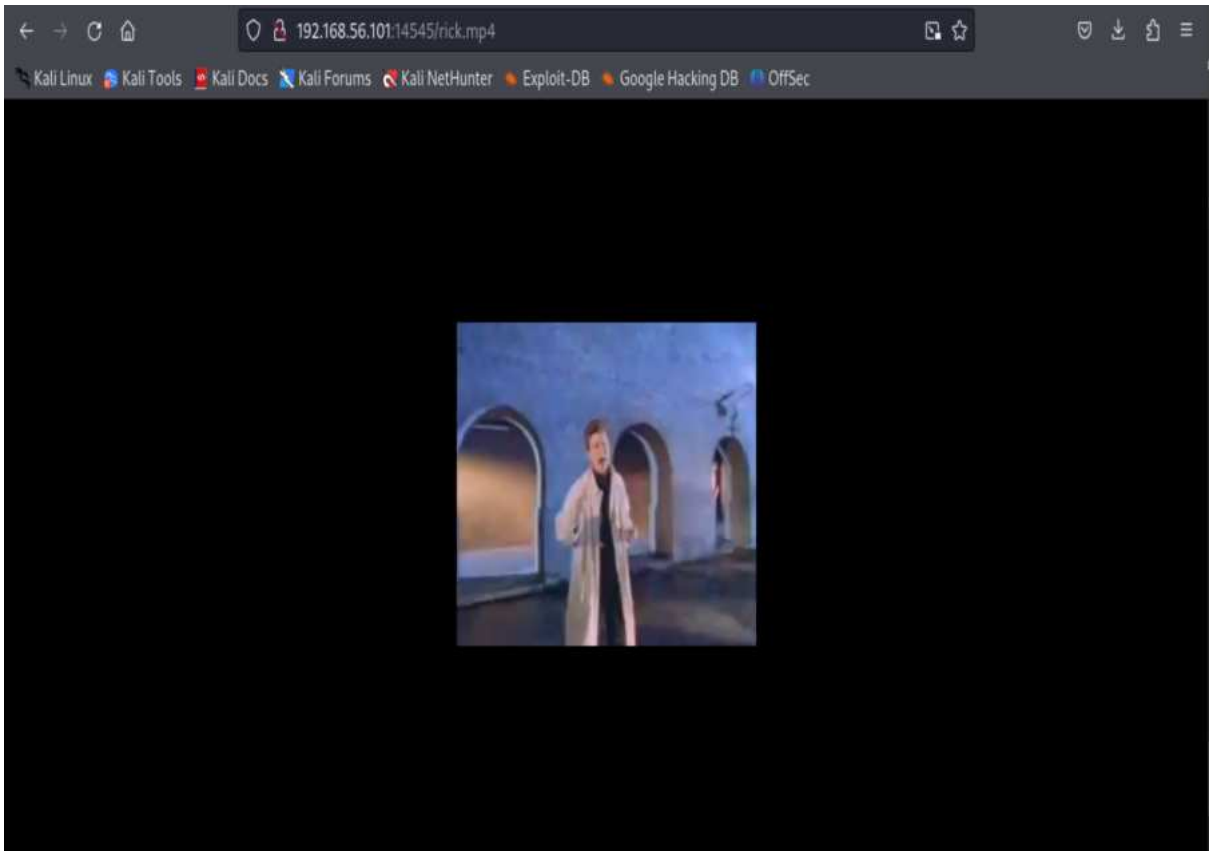
- We have used the command **"nmap -p- 192.168.56.101" (image 29).**
- There are a **total of 65,535 TCP ports in a system**, but nmap only scans for the **common 1000 ports** this would lead to missing hidden ports like in our case**.**

- So, -p- means full port scan which scans all **65,535 ports** gathering all the hidden ports too.
- As displayed in image 29, full port scan was done.
- We were mentioned that the server is open on a different port and not the default port 80.
- According to the scan results in image 29, there is an unknown service running on **TCP port 14545.**
- Now we accessed the port in our browser.

[image 30 – port 14545]

- As displayed in image 30, we accessed the port 14545 in our Linux browser.
- http://192.168.56.101:14545
- This opened a separate webpage different from the main page.
- The webpage mentioned about Configuration files, Where the DocumentRoot is located and information about CGI Scripts.

- **It also suggested us to check the source code of the webpage and had a link with the name "RICK" near it.**



[image 31 – RickRolled]

- The link played a video song by rick Astley "Never Gonna Give You Up" (image 31).
- Next, we moved on to inspect the source code of the webpage by using **"Ctrl+U"** in the browser.
- This opened the HTML code of the webpage and by using the find shortcut, we searched for **"Flag"** as it has been the common name for the files of previous three flags.

```
1 <!DOCTYPE html PUBLIC "-//W3C//DTD XHTML 1.0 Transitional//EN" "http://www.w3.org/TR/xhtml1/DTD/xhtml1-transitional.dtd">
2 <html xmlns="http://www.w3.org/1999/xhtml">
3 <head>
4 <meta http-equiv="Content-Type" content="text/html; charset=UTF-8" />
5 <title>Welcome page</title>
6 <style type="text/css" media="screen">
7 body { background: #e7e7e7; font-family: Verdana, sans-serif; font-size: 11pt; }
8 #page { background: #ffffff; margin: 50px; border: 2px solid #c0c0c0; padding: 10px; }
9 #header { background: #4b6983; border: 2px solid #7590ae; text-align: center; padding: 10px; color: #ffffff; }
10 #header h1 { color: #ffffff; }
11 #body { padding: 10px; }
12 span.tt { font-family: monospace; }
13 span.bold { font-weight: bold; }
14 a:link { text-decoration: none; font-weight: bold; color: #C00; background: #ffc; }
15 a:visited { text-decoration: none; font-weight: bold; color: #999; background: #ffc; }
16 a:active { text-decoration: none; font-weight: bold; color: #F00; background: #FC0; }
17 a:hover { text-decoration: none; color: #C00; background: #FC0; }
18
19 body {
20   background-image: url('locard.jpg');
21   background-repeat: no-repeat;
22   background-attachment: fixed;
23   background-size: 100% 100%;
24 }
25
26 </style>
27 </head>
28 <body>
29 <div id="page">
30  <div id="header">
31  <h1> The server that does not have anything to "serve" </h1>
32   This is an empty server for the most part.
33   You might think that this is a simple diversion from finding the flag...
34   If you search you will find the flag
35  </div>
flag   Highlight All   Match Case   Match Diacritics   Whole Words   1 of 4 matches   X
```

[image 32 – Source code]

```
45
46  <p>
47   Maybe check the source code?
48   <a href="rick.mp4">Rick</a>
49  </p>
50 </div>
51 </div>
52 <!-- COMMENT -->
53 </body>
54 <!--FLAG 4
55 Well done you have found the FLAG
56 it is:
57  88
58  -->
59 </html>
60
61
```

[image 33 – Flag 4 Result]

- As expected, by using the search option we were able to easily find the Flag successfully (image 33).
- With this, the fourth flag "**88**" was found and documented successfully (image 33)

8. Flag 5 – Logging into DF live server for the Final Flag:

- This is the final flag.

Flag 5

You need to gain root access to find flag 5

There are lots of ways of doing this but if you have flag 1 to 4 you should join them together...

That is, the root password consists of 8 numbers

[image 34 – Flag 5]

- In this we are given a task to access the DF live server in order to find the fifth flag (image 34).
- It is also mentioned that the password for the server is the combination of the first four flags (image 34).
- The username was already mentioned as **"root"** in the beginning (image 34).
- So, we have used
    - Name : root
    - Password : 22446688

```
digitalforensics1 login: root
Password:
Last login: Mon Nov 10 16:37:41 UTC 2025 from 192.168.56.1 on pts/0
Welcome to Ubuntu 18.04.6 LTS (GNU/Linux 4.15.0-213-generic x86_64)

 * Logging the user to the server...


Thu Nov 27 20:48:35 UTC 2025

*****************************************************************
          Digital Forensics Training Lab 0.5

Version 2
                                                PC
*****************************************************************

The IP address for this machine is:

        inet 192.168.56.101  netmask 255.255.255.0  broadcast 192.168.56.255

root@digitalforensics1:~# _
```

[image 35 – DF login]

- Using the Credentials we have successfully logged in into DF server (image 35).
- Like we did in the FTP server, we are listing the files in the server.

```
root@digitalforensics1:~# ls
hint.txt  subfolder
root@digitalforensics1:~# cat hint.txt
```

[image 36 – listing files in DF server]

- The server had 2 files **hint.txt and solder named subfolder (**image 36**).**
- We decided to get the contents in the hint.txt using **"cat hint.txt"**

```
root@digitalforensics1:~# ls
hint.txt  subfolder
root@digitalforensics1:~# cat hint.txt
...

Excellent my friend!

You have gained root access...
The force is strong with you!

The final flag is in a subfolder
With your skills you should have absolutely no problem finding it

...


root@digitalforensics1:~#
```

[image 37 – hint.txt**]**

- The **hint.txt** file was extremely useful as it directly displayed the final location of the flag 5.
- Flag 5 is in the **"subfolder"** directory.
- Now we are going into the subfolder directory using **"cd subfolder".**

```
root@digitalforensics1:~# cd subfolder
root@digitalforensics1:~/subfolder# ls
FLAG  theresNoOtherFilesHereToConfuseYou.txt
root@digitalforensics1:~/subfolder# cat FLAG
######

............

FLAG 5

You have discovered the final flag

Well done

You have made it to the end successfully

............

The full answer is:

22 44 66 88 99
root@digitalforensics1:~/subfolder#
```

[image 38 – Flag 5 Result]

- The subfolder directory had 2 files.
- **"FLAG",** and another file with a large name (image 38).
- But obviously the file name with **FLAG** would have the FLAG 5 (image 38).
- Like expected while seeing the contents in the **FLAG** file we were able to find the final flag (image 38).
- With this, the final flag "**22 44 66 88 99**" was found and documented successfully (image 38).

9. Flag Summary:

Flag 1 = 22

Flag 2 = 44

Flag 3 = 66

Flag 4 = 88

Flag 5 = 22 44 66 88 99

9. Other Details:

FTP server login:

Username : test

Password : test

10. Commands Used:

1. **ifconfig**

2. **echo -n 22 | md5sum**

   echo -n 22 | sha1sum

3. **nmap 192.168.56.101**
4. **nmap -p- 192.168.56.101**
5. **ftp 192.168.56.101**
6. **ls**
7. **cd directory**
8. **cd files**
9. **get FLAG**
10. **get FLAG2.txt**
11. **get info.info**
12. **get readme.txt**
13. **get hint.txt**
14. **bye**
15. **cat FLAG**
16. **cat FLAG2.txt**
17. **cat info.info**
18. **cat readme.txt**
19. **cat hint.txt**
20. **dirb http://192.168.56.101/**
21. **cat hint.txt**
22. **cd subfolder**
23. **cat FLAG**

11. CONCLUSION

- In this training, I basically worked through the system step by step and used different forensic methods to find all five flags in the DF Training virtual machine. I started by setting up both virtual machines, making sure the IP addresses matched, and checking that Kali Linux could communicate with the DF Training server. Once everything was connected properly, I opened the server in the browser and understood how the challenge was structured.
- For Flag 1, I used hashing commands with MD5 and SHA-1 to match the given values and find the correct two-digit number. After several tries, the number 22 matched both the hashes, so that became the first flag. For Flag 2, I did an Nmap scan and found that the FTP port was open. I logged in with the simple username and password ("test") and looked through all the directories. One folder had a fake flag, but the real flag was inside the files directory in a file called FLAG2.txt.
- For Flag 3, the hint told me to use a directory-brute-forcing tool, so I ran DIRB with the IP and HTTP. DIRB showed me many hidden folders, and by checking them one by one, I found a path that contained the flag3.txt file. That gave me the third flag. Then for Flag 4, I scanned all ports using Nmap -p- and found a strange port, 14545, running an HTTP service. When I opened it, it showed a random page with a Rickroll video. Since there was no flag on the page, I viewed the page source and found the fourth flag hidden in an HTML comment.
- For Flag 5, I combined all four flags (22, 44, 66, 88) to form the root password: 22446688. I logged into the DF Training machine as root and checked the files inside the home directory. The hint told me to look in the subfolder, and inside that folder I found the last flag written in a file called FLAG.
- Overall, this training helped me understand how to move through a system logically and check each service properly. I used hashing, FTP, directory scanning, port scanning, and root access—all in one task. It showed me how small clues connect together and how to carefully check each part of a machine. Completing all the flags helped me get more comfortable with basic forensic steps and how to investigate a system from start to finish.